

# Training and Testing

Project: Transparent Book Recommender

## 1. Training and Testing

### Teach the learning cycle

Primary target: Choose a split unit that matches the intended prediction claim

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/11, ai-4/8. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

## Retrieval answer guidance

### Human Classifier Simulation

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

Look for: Construct a classifier with a stated input format, selected rule and recorded output

Reference example, not the only acceptable wording: Require a present height, unit cm, a whole number and range 1 through 10 inclusive. For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range. Fractional values such as 4.5 cm also fail. The contract defines accepted inputs; it does not list only the fitting records or establish prediction correctness.

### Operational Labels

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a category definition using explicit evidence and boundary rules

Reference example, not the only acceptable wording: In order: ready because both known fields pass; revise because 31 exceeds 30; review because the known day passes but the room is unknown; revise because the known day failure has priority over the unknown room. Equality at 30 is accepted.

### Loops and Stopping

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased unseen test result. Explain the issue and revise the evaluation plan.

### Concept to explain

The test set estimates performance on unseen cases.

#### Term · Meaning

**Term:** training set

**Meaning:** Examples used to adjust a model's parameters or fit learned preprocessing.

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**Term:** test set

**Meaning:** Data reserved for evaluating a selected system; final test data must not influence its development.

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**Term:** unseen

**Meaning:** Not previously used for the development, practice or selection being evaluated.

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**Term:** leakage

**Meaning:** Use of information during development or prediction that should be unavailable for a fair evaluation of the intended task.

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## Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

## Preparation for the connected explanation

Teach two 45-minute sessions with fictional frame cards. Focus on the evaluation goal and unit of separation before percentages. The worked plan separates trips for an intended new-trip claim on a familiar route. It neither trains a model nor establishes performance. Distinguish frames, groups and sets, and retain Week 5 label-rule and review records.

## Readiness answer

They can share the same mistake or use an unclear rule. Agreement counts matching labels; correctness requires checking the rule and evidence.

Printed lesson and workbook

Twelve frame cards A1 A2 A3 B1 B2 C1 D1 D2 E1 E2 F1 F2

Three trays or paper areas named training validation and test

Optional Python 3 editor with no extra packages

## End goal for the pictorial case

By the end, I can split related records by group when evaluation concerns new groups.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## 2. Explain the mechanism

### Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. 20 cards: seal 5 as test cases first.
2. Design the rule using only the remaining 15.
3. Freeze the rule before opening the test cards.
4. If a test label guided a change, it is no longer a fair final test.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### Concrete teaching case

A rule is developed and evaluated using the same books A–H.

Why does this not establish unseen performance?

### Exact explanation to model

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

### Explain the added worked example

A model may seem successful when its test examples closely resemble examples it already learned from. Imagine several pictures taken seconds apart during one robot trip. They have different picture IDs but share a route, lighting and objects. This week you will assign entire trips to separate sets and explain what the resulting evaluation could measure. No model is fitted in this lesson.

Our invented twelve-row dataset contains frame descriptions from six trips. Each row is one frame; the group is its trip ID. The proposed claim concerns new trips on the same practice route under similar conditions. We therefore keep all frames from a trip in one set. To claim success on new routes, grouping

only by trip would not be enough.

This small fictional assignment is for learning split design, not evidence of real performance. Each frame belongs to one group and one set. Store frame and trip IDs as metadata for auditing; do not assume an ID is an appropriate prediction input.

Training data fits the model. Validation data helps compare choices such as candidate rules. Keep the test set for a final check after those choices are fixed. A learner who repeatedly changes the model after seeing test results has used that set for development; its score is no longer an untouched final check.

The first line of output is []: no trip ID is shared between any pair of sets. The second is 3 1 2: training, validation and test contain three, one and two trip groups. The symbol & finds shared values and | combines sets. These are group counts, not frame counts.

This audit assumes the lists correctly describe every frame assignment. It cannot find an omitted frame, a wrong trip ID, a copied image with a different ID or a feature containing the answer. Check the assignment table and source records too.

## Key terms

### Term · Meaning

**Term:** Training set

**Meaning:** Examples used to fit a model

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**Term:** Validation set

**Meaning:** Separate examples used to compare choices during development

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**Term:** Test set

**Meaning:** Held out examples used for final evaluation after choices are fixed

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**Term:** Group

**Meaning:** Related examples that must stay together for a stated evaluation goal

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## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Suppose A1 and A2 are in training while A3 is in test. The frame IDs differ, but A appears in both sets.

Shuffling the rows does not remove this relationship. For our new-trip claim, move all A frames to one chosen role and record the revised assignment before evaluating a model. Do not copy A3 into both roles.

Audit that every source frame occurs exactly once, no group crosses roles and the assigned counts sum to the source count. In our worked plan the row counts are six training, two validation and four test: twelve total. A fixed random seed can reproduce a random split, but it cannot make a flawed grouping choice fair.

**A learner moves D2 from validation to training but leaves D1 in validation. Give the new row counts and shared trip ID. Explain why different frame IDs and a fixed seed do not repair the overlap.**

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A split record names the prediction goal, source version, group definition, exact assignments, row and group counts, date and any random method or seed used. For this hand-assigned example, record that there was no random sampling. Keep label-rule versions and unresolved review statuses traceable as in Week 5.

**Thinking move · Teacher prompt**

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

**A pictorial worked case: Training and Testing**

A fictional catalogue model learns from scans of book covers and will face previously unseen books.

## The facts we will use

Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed training A1,B1,C1; validation A2,D1; test B2,C2,D2.

AI ACADEMY / AI-2/11

## Training and Testing

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Choose unit

The intended kind of new example being evaluated is a new book, not merely a new scan.



#### 2. Find leakage

A crosses train/validation; B,C cross train/test; D crosses validation/test.



#### 3. Group repair

Training all A,B scans; validation all C scans; test all D scans.



#### 4. Freeze assignment

Keep every scan from one book in exactly one role before fitting.

### CHANGE ONE THING

Give A2 a new filename without changing the book it depicts.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Choose unit

**What happens and why:** The intended kind of new example being evaluated is a new book, not merely a

new scan.

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**Step:** 2. Find leakage

**What happens and why:** A crosses train/validation; B,C cross train/test; D crosses validation/test.

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**Step:** 3. Group repair

**What happens and why:** Training all A,B scans; validation all C scans; test all D scans.

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**Step:** 4. Freeze assignment

**What happens and why:** Keep every scan from one book in exactly one role before fitting.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. Grouping by book matches the intended claim about new books and makes the split inspectable. The repaired split has four training, two validation and two test scans, but only two, one and one books respectively.

### What this example does not establish

One validation book and one test book provide very limited diversity even though no groups overlap.

### Change one thing in the pictured case

Change: Give A2 a new filename without changing the book it depicts.

Prediction: Its group is still A; the original related examples still cross the split.

Reason: Renaming a row does not remove the relationship between scans.

### Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

## 3. Demonstrate and guide practice

### Demonstrate and guide practice

## Board sequence

Write the given facts: A rule is developed and evaluated using the same books A–H.

Write the question to explain: Why does this not establish unseen performance?

Reveal after predictions: Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

Keep for an independent attempt: You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test evidence?

## Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

## Model these guided questions

Guided: the row counts become 7 training, 1 validation and 4 test. D is shared by training and validation because D1 and D2 are related frames from one trip. Unique frame IDs and a seed do not remove the shared trip. Return D2 to validation to recover the worked plan. Exit: separate trip IDs on one route do not test unseen routes. Hold out whole routes with all their trips, and collect enough routes and relevant conditions for the intended evaluation.

## Case-specific teaching sequence

### Stage · Teacher action

**Stage:** Session 1 start to 7 minutes

**Teacher action:** Retrieve why matching labels can share an error. State the new-trip evaluation goal and define one row and one group.

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**Stage:** 7 to 18 minutes

**Teacher action:** Use the frame cards to build the worked split. Count rows and groups separately and explain the three roles.

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**Stage:** 18 to 32 minutes

**Teacher action:** Trace the intersection audit. Complete W1 and W2; ask which metadata problems an empty overlap list could miss.

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**Stage:** 32 to 45 minutes

**Teacher action:** Move D2 into training and complete the guided question. Repair it by moving it back, then record the before and after counts.

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**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve why shuffling and fixed seeds do not remove relationships. Ask what must stay together for a new-route claim.

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**Stage:** 8 to 20 minutes

**Teacher action:** Complete W3 and discuss group-based, route-based and time-respecting evaluation goals. Explain why percentages cannot decide the goal.

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**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently before hints. Require all three overlap checks, exact repaired frame lists and both count units.

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**Stage:** 35 to 45 minutes

**Teacher action:** Discuss W5 and the exit question. Use the fresh retry after support and record what help was needed.

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## Reduce help deliberately

Completion task: For the worked A through F plan, give training, validation and test counts first in groups and then in rows. Interpret [] from the code and state one problem it cannot detect.

Teacher check: Group counts are 3,1,2; row counts are 6,2,4. The empty list means no trip ID overlaps between any pair of the listed sets. The code cannot detect wrong metadata, an omitted frame, a disguised duplicate or an outcome feature.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Different frames may share a trip or source. Choose group metadata based on the claim and verify it against the source.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictorial practice question

How many distinct books are in each repaired role, and why report that as well as scan counts?

## Reasoned teacher answer

Training 2, validation 1, test 1. Scan counts alone exaggerate the number of distinct books evaluated.

Changed-case reasoning: Its group is still A; the original related examples still cross the split. Renaming a row does not remove the relationship between scans.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

# 4. Run the weekly evidence lab

## Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Split cards before rule-making. Record the split, build on training only, freeze the rule, then reveal and score the test pile once.

What did the hardest test teach you that the normal test did not?

## Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

## Record · What to capture

**Record:** Input or evidence

**What to capture:** Exact data, instruction or claim

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**Record:** Expected result

**What to capture:** Answer or safe fallback declared before the run

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**Record:** Observed result

**What to capture:** Actual result, including errors

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**Record:** Explanation

**What to capture:** Rule, calculation or evidence explaining the outcome

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**Record:** Next test

**What to capture:** One justified change and a new prediction

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## Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

## Address these additional misconceptions

Different row IDs mean independent examples: Different frames may share a trip or source. Choose group metadata based on the claim and verify it against the source.

A fixed percentage guarantees a fair test: Percentages do not remove overlap or ensure enough groups and relevant cases. The goal and dependencies determine the split.

An empty overlap list proves there is no leakage: The audit checks listed group IDs only. Wrong metadata, copied content, outcome features or preprocessing fitted on all sets can still invalidate evaluation.

## Match the support to the first error

### What the response shows · Next teaching action

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

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**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Assess and explain the answers

#### Worked case answer

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

#### Independent question

You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test evidence?

#### Expected independent answer

No. Its result influenced development. Use a separate untouched test after tuning; keep the earlier failure in the development record.

## What counts as evidence

**Criterion · 0 points · 1 point · 2 points**

**Criterion:** Result

**0 points:** Incorrect or absent

**1 point:** Partly correct

**2 points:** Correct with necessary conditions

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**Criterion:** Reasoning

**0 points:** No supporting trace

**1 point:** Some steps

**2 points:** Clear mechanism, calculation or source support

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**Criterion:** Limit

**0 points:** Overclaims certainty

**1 point:** Vague limitation

**2 points:** Relevant uncertainty or missing fact

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**Criterion:** Transfer

**0 points:** Copies without adapting

**1 point:** Adapts with prompts

**2 points:** Solves a changed case independently

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Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

## Feedback

Ask the learner to find the evidence that challenges this claim: “A file named test can never become development data.” Keep the first attempt and an uncoached retry separate.

## Original transfer answer

The ten books are validation data because their scores selected the version. Evaluate the chosen rule once on a fresh labelled final set that did not influence any choice. Disclose small sample size and any related-book overlap.

## Explanations for the additional workbook examples

### W1 Explain the three roles

Explain what training, validation and test sets are used for. What changes if you repeatedly choose a model using its test score?

Training fits the model; validation compares development choices; test evaluates after choices are fixed. Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

### W2 Distinguish frames from groups

For the worked A through F plan, give training, validation and test counts first in groups and then in rows. Interpret [] from the code and state one problem it cannot detect.

Group counts are 3,1,2; row counts are 6,2,4. The empty list means no trip ID overlaps between any pair of the listed sets. The code cannot detect wrong metadata, an omitted frame, a disguised duplicate or an outcome feature.

### W3 Repair a related example leak

A1 and A2 are in training and A3 is in test. Explain why unique frame IDs and a fixed random seed do not fix this. Propose a repair and what to record.

The frames share trip A, so the test contains a trip already represented in training. Unique frame IDs do not remove similarity; a seed only reproduces the assignment. Put every A frame in one role, keep each frame once and record the revised assignments and counts before model evaluation.

### W4 Audit and repair independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is training P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared groups for each pair of sets. Repair it by assigning P and Q to training, R to validation and S to test. List all frames in each role and give row and group counts. State one limit of the repaired plan.

Training/validation share P; training/test share P and Q; validation/test share P and S. Repaired training is P1 P2 P3 Q1 Q2 (5 rows, 2 groups), validation R1 (1 row, 1 group), test S1 S2 (2 rows, 1 group). Every frame appears once and groups are disjoint. This tiny plan does not establish reliable performance, correct metadata, label coverage or success on new routes.

### W5 Match separation to the claim

A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative. Is this enough for a strong performance claim?

Both routes occur in training and test, so the plan does not hold out a new route. For illustration, keep A and B together for training and C and D together for test, or reverse the route roles. With only two routes there is too little evidence for a strong general claim and no separate route left for validation; obtain more routes for a fuller design.

## Assess independent transfer

Selected case: Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is training P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared groups for each pair of sets. Repair it by assigning P and Q to training, R to validation and S to test. List all frames in each role and give row and group counts. State one limit of the repaired plan.

## Selected case answer

Training/validation share P; training/test share P and Q; validation/test share P and S. Repaired training is P1 P2 P3 Q1 Q2 (5 rows, 2 groups), validation R1 (1 row, 1 group), test S1 S2 (2 rows, 1 group). Every frame appears once and groups are disjoint. This tiny plan does not establish reliable performance, correct metadata, label coverage or success on new routes.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

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Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

## Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## 6. Connect and extend learning

### Connect and extend learning

#### Transparent Book Recommender

Keep final user profiles hidden until weights and thresholds are fixed.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

#### Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

#### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

#### Bridge to the next week

Next: Week 12: Accuracy as a Fraction. Retrieve today's evidence and identify what carries forward and what is new.

#### Support and fresh reassessment

Keep a coloured border shared by cards from the same trip. Supply blank assignment tables without the correct roles. Extension: explain why a new-route evaluation with many frames but only two routes has limited evidence and cannot obtain three nonempty disjoint route sets.

Require W4 overlaps P; P and Q; P and S, in pair order, then exact repaired lists with rows 5,1,2 and groups 2,1,1. Fresh retry: U1 U2 belong to U, V1 to V, W1 W2 to W. Proposed training U1 V1, validation W1, test U2 W2 has shared U between training/test and W between validation/test; training/validation share none. Repair to all U in training, V in validation and all W in test: rows 2,1,2 and groups 1,1,1. Require an explicit small-sample limitation.

Week 7 introduces a baseline before a model. Carry forward the split record and keep the final test protected while comparing choices. Teacher reference: scikit-learn cross-validation documentation, grouped data section, [https://scikit-learn.org/stable/modules/cross\\_validation.html](https://scikit-learn.org/stable/modules/cross_validation.html) (reviewed 2026-09-27).

#### Plan the delayed return

Suggested return points in this level: ai-2/12, ai-2/14, ai-2/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

## 7. Prepare, teach and assess

Teaching target: Choose a split unit that matches the intended prediction claim

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

### Session · Plan · Resources

**Session:** Session 1 · Understand · 45 min

**Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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Before class, solve the supported practice: For the worked A through F plan, give training, validation and test counts first in groups and then in rows. Interpret [] from the code and state one problem it cannot detect.

Reasoned practice answer: Group counts are 3,1,2; row counts are 6,2,4. The empty list means no trip ID overlaps between any pair of the listed sets. The code cannot detect wrong metadata, an omitted frame, a disguised duplicate or an outcome feature.

Check every learner before discussion: Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Different frames may share a trip or source. Choose group metadata based on the claim and verify it against the source.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Give A2 a new filename without changing the book it depicts. Require a prediction and a reason before comparing. Expected result: Its group is still A; the original related examples still cross the split.

Reason: Renaming a row does not remove the relationship between scans.

Independent assessment: withhold the reserved answer until the learner has attempted the new case.

Record support separately from correctness.

### **Criterion · Not yet · With support · Independent**

**Criterion:** Explanation

**Not yet:** Names an answer without a deciding fact

**With support:** Uses a hint to link fact and decision

**Independent:** Explains the deciding fact in own words

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**Criterion:** Trace or test

**Not yet:** Cannot connect input and result

**With support:** Completes steps with prompts

**Independent:** Traces or tests a changed case accurately

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**Criterion:** Limits and revision

**Not yet:** Claims more than evidence supports

**With support:** Names a limit after a prompt

**Independent:** States a defensible limit and useful next test

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Project checkpoint: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

## **8. Use feedback, labs and repair paths**

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved

independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.